

Contracting, Non-Normal, and Not the Mechanism: A Pre-Registered Spectral Audit of Latent Chain-of-Thought Reasoning

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Abstract

A Coconut-style latent reasoner feeds its last hidden state back as the next input embedding, so its chain of continuous thoughts is literally a discrete-time dynamical system $c_{t+1} = F_t(c_t)$. We pre-registered — with kill conditions committed before any result — the natural dynamical-systems hypothesis for the emerging field of latent-reasoning interpretability: that the spectral invariants of this map (transient-expansion and slow-mode structure of the step Jacobians, and orbit-level Koopman estimates) *predict* the established anchor/branch structure of the reasoning, validated against ProsQA’s ground-truth DAG. The hypothesis fails its controls, informatively. The step maps are strongly contracting (mean spectral radius 0.34) yet highly non-normal (mean $\sigma_1 = 2.97$, σ_1/ρ ratio 8.96): thoughts are transiently amplified in singular directions while every eigenmode decays. But every spectral invariant’s apparent branch/anchor signal survives in pause-token models with *no recurrence at all*, does not respond to surgically de-branching the problems, and collapses to chance — along with all trivial baselines — under within-step stratification: the observable spectral structure is architectural and positional, not reasoning-specific (kill condition K1). Causally, perturbations along the top singular direction of the step Jacobian produce $3.32\text{--}7.03\times$ the downstream effect of magnitude-matched random directions (eigendirections carry no privilege — the non-normal signature), yet the absolute effects are minute: replacing any single thought outright shifts the answer margin by a median of 0.002 against baseline margins of ≈ 13.9 , flipping zero of 500×6 ablations — and transplanting *all six* thoughts from a random donor problem changes zero of 500 answers. Influence Jacobians differentiated through all paths show why: within three steps, about half of a thought’s influence flows through the attention cache rather than the recurrence (median routedness up to 0.51), and a pooled orbit-level operator inherits the same positional confounds. For this high-accuracy checkpoint, the recycled thought vector is a weak causal channel whose spectral geometry says little about the reasoning it hosts — a pre-registered cautionary result extending “observable patterns are not explanations” to the spectral/dynamical domain, with matched controls and causal validation as the non-negotiable price of admission for state-space interpretability of latent reasoning.

1 Introduction

A latent (continuous) chain-of-thought model of the Coconut family [1] reasons by feeding its own last hidden state back to itself as the next input embedding. The result is a trajectory of continuous “thought” vectors $c_1, \dots, c_n \in \mathbb{R}^d$, generated by a deterministic autoregressive map

$$c_{t+1} = F_t(c_t), \tag{1}$$

where F_t is the transformer evaluated at the t -th latent slot with the problem context held in the attention cache. Equation (1) is a discrete-time dynamical system, and this paper asks the dynamical-systems question about it: do the *spectral invariants* of F_t — the eigenvalues and singular structure of its local Jacobian $J_t = \partial c_{t+1}/\partial c_t$, and the spectrum of orbit-level transfer-operator estimates — *predict* the interpretable structure of the reasoning that the trajectory implements?

Two pieces of that interpretable structure are already established by prior causal work, and we treat them as background rather than findings. First, latent reasoning steps have heterogeneous, outsized causal influence: some thoughts are *anchors* whose perturbation reroutes the computation, while others are nearly inert [2]. Second, the causal load of latent steps concentrates in *low-rank directions* of representation space, and observable geometric patterns alone — decodable frontiers, structured step geometry — appear equally in matched controls that lack the latent-reasoning mechanism, so patterns cannot be read as mechanism without causal tests [3].

What is missing is a *predictive, mechanistic account*: a quantity computable from the dynamics alone — with no access to labels, probes, or interventions — that says *which* steps will be anchors and *where* the reasoning branches, and *why*. Dynamical systems theory supplies natural candidates. A reasoning step at which the model must select among several admissible continuations is, dynamically, a step of *transient expansion*: locally, nearby thought states must separate, which is the signature of a large leading singular value $\sigma_1(J_t)$ — possibly without any eigenvalue of large modulus, i.e. *non-normal* amplification [4, 5]. A thought whose content must persist and steer several subsequent steps is, dynamically, aligned with *slow modes*: eigenvalues of J_t near the unit circle, the discrete-time analogue of marginal stability, and the quantity that Koopman-spectral diagnostics of transformers have found most informative on other axes [6].

Contributions. This paper reports what happened when that program was run properly — pre-registered, with kill conditions committed before results, matched behavioral controls, and causal validation:

1. **The first spectral characterization of the latent-thought autoregressive map.** The step Jacobians of a 98%-accurate Coconut/ProsQA model are strongly contracting (mean $\rho = 0.34$) yet highly non-normal (mean $\sigma_1 = 2.97$, $\kappa = 8.96$, normalized Henrici departure 0.96): transient amplification without eigenvalue growth. The thought vector itself rides the dominant eigen-subspace of its own linearization (90.6% of its norm in the top-4 modes).
2. **A clean pre-registered negative on spectral prediction of reasoning structure (K1).** Branch/anchor prediction from spectral invariants fails all three matched controls: pause-token models without any recurrence show equal or larger “branch signatures”; paired pruned twins with the branching surgically removed show unchanged spectra; and within-step stratification collapses every predictor — including trivial trajectory baselines — to chance. The observable spectral structure is a property of the architecture, the encoding, and the step position.
3. **A directional causal finding with an honest ceiling.** The residual per-thought causal signal concentrates in the *non-normal amplification directions* $v_1(J_t)$ ($3.32\text{--}7.03\times$ random at matched magnitude; eigendirections carry no privilege) — a dynamical identity for the low-rank causal directions of Aswal et al. [3] — but the channel itself is nearly inert: no single-thought intervention of any kind flipped a single answer, and a post-hoc probe replacing all six thoughts at once (slot means, zeros, or a random donor problem’s entire chain) leaves accuracy at baseline. The recycled content is collectively unused; the latent segment contributes structure, not state.

4. **A quantitative bound on state-space accounts of latent reasoning.** Differentiating the fully unrolled map shows the chained local Jacobians lose about half of a thought’s influence within three steps (median routedness 0.51 at $t=1, k=3$); the influence is routed through the attention cache, in operator-norm agreement with the interventional routing reported by Li et al. [2], and pooled orbit-level (EDMD/Koopman) estimates do not recover reasoning-specific structure either.

Distinguishing statement. Prior work establishes that latent reasoning has heterogeneous, causally-routed anchor structure [2] and that low-rank directions carry its causal load [3]; separately, Jacobian- and Koopman-spectral tools have been applied to the transformer’s *depth* axis — treating layers as time on standard forward passes [6, 7]. We are, to our knowledge, the first to characterize the latent-reasoning *autoregressive* map $c_{t+1} = F_t(c_t)$ spectrally, and to test whether its dynamical invariants — dominant/slow modes and non-normal transient-expansion directions — *predict* the anchor thoughts and branch points of the reasoning, validated against a ground-truth reasoning DAG (ProsQA) with matched pause-token and pruned linear-chain null controls, and checked *causally* by intervening along the spectrally identified directions. We adopt the discipline of Aswal et al. [3] throughout: no spectral pattern is interpreted as mechanism unless it survives the matched controls and the interventions.

2 Background and substrate

Coconut and ProsQA. Coconut [1] trains a language model to replace explicit chain-of-thought tokens with latent thoughts: at a designated latent slot, the input embedding is the last hidden state of the preceding position rather than a token embedding. Training uses a curriculum that replaces reasoning steps with latent steps stage by stage. ProsQA [1] is a synthetic graph-reachability QA benchmark: each problem lists edge statements (“Every gerpus is a terpus.”, “Sally is a boomus.”) over nonsense concepts, then asks which of two candidate concepts a named individual belongs to. Crucially for us, each problem ships its full DAG, the ground-truth reasoning path $v_0 = \text{root}, v_1, \dots, v_L = \text{target}$ ($L \in [3, 6]$), and the distractor answer, so “where the reasoning branches” is known exactly: we label step t a *branch step* iff the out-degree of v_{t-1} exceeds one — a real choice existed when leaving v_{t-1} . In the test split, 81.7% of on-path steps are branch steps.

Checkpoints and matched controls. We analyze the public `coconut-curriculum-checkpoints` suite [8]: GPT-2 (124M) [9] fine-tuned on ProsQA with the 7-stage Coconut curriculum, in four variants. M2 is Coconut proper (hidden-state recycling; 97.0% test accuracy as published). M3 and M4 are *pause-token* controls [10]: identical curriculum, identical latent-slot positions, but a learned pause embedding is fed at every latent slot — no recycling, hence no recurrence — and they match Coconut’s accuracy (96.6% / 94.8%). M1 is the standard chain-of-thought baseline (83.0%). The pause models are the decisive null: same training distribution, same positions, same behavior, different mechanism [3, 8]. Our harness reproduces the published accuracies (M2 98.0%, M3 95.4%, M4 95.0%, M1 76.8% under our decode protocol).

The trajectory and its step maps. Evaluation feeds question $\| \langle \text{start-latent} \rangle \| 6 \times \langle \text{latent} \rangle \| \langle \text{end-latent} \rangle$ and decodes the answer. We write c_t ($t = 1..6$) for the vector fed at latent slot t ; c_1 is the hidden state at the start-latent position, and for M2, c_{t+1} is the hidden state at slot t . The local step map F_t holds the prefix attention cache fixed, so $J_t = \partial c_{t+1} / \partial c_t \in \mathbb{R}^{768 \times 768}$ is exact (computed by automatic differentiation; we verify it against finite differences). For the pause models the *identical*

architectural object exists at the identical positions — the Jacobian of the slot map evaluated at the pause embedding — and any spectral structure appearing equally there is a property of the architecture and the problem encoding, not of the latent recurrence. Note that c_t influences later steps through two routes: the direct feed (1) and the attention cache written at slot t ; the local Jacobian captures only the first, which is exactly the gap probed in Section 6.

3 Methods

Spectral invariants. For each problem and step we eigendecompose and SVD-decompose J_t and record: spectral radius ρ ; leading singular values σ_1, σ_2 and the count of expanding singular directions $\#\{\sigma_i > 1\}$; the Henrici departure from normality $d_F(J) = \sqrt{\|J\|_F^2 - \sum_i |\lambda_i|^2}$ [11], raw and normalized by $\|J\|_F$; the non-normal amplification ratio $\kappa = \sigma_1/\rho$; and the *near-unit spectral mass* $\sum_{|\lambda_i| \in [0.9, 1.1]} |\lambda_i|$, adapted from the training-stability diagnostic of Kim et al. [6] to the thought axis. Per pre-registration (PLAN.md, committed before results), the *primary* branch predictor is σ_1 (transient expansion) and the primary anchor predictor is the near-unit spectral mass; all other invariants are secondary. Every predictor is benchmarked against three trivial baselines: the step index t , $\|c_t\|$, and $\|c_{t+1} - c_t\|$.

Ground truth. Branch labels come from the ProsQA DAG as above (padding steps beyond the path length are excluded). Anchor ground truth is measured by our own step ablations, following the leverage protocol of Li et al. [2]: replace c_t with the dataset-mean thought for slot t (robustness: the zero vector), recompute the downstream recurrence, and record the influence $I_t = |\Delta \text{margin}|$, where margin is the teacher-forced log-probability difference between the correct and distractor answers.

Statistics. All discrimination results are pooled AUROC / average precision with 95% cluster-bootstrap confidence intervals (resampling problems, 10^4 resamples); rank relations are Spearman correlations with the same bootstrap; paired model-vs-control comparisons use Wilcoxon signed-rank tests. Kill conditions K1–K3 (null-control failure, causal failure, routed-effect failure) were pre-registered in PLAN.md and committed to the repository before any result existed.

4 E0: do spectral invariants predict the DAG?

What the step maps look like. The latent step maps of the trained Coconut model are *strongly contracting yet highly non-normal*: across 500 problems and six steps, the mean spectral radius of J_t is 0.34 while the mean leading singular value is 2.97 ($\kappa = \sigma_1/\rho$ averages 8.96), and the normalized Henrici departure from normality averages 0.96 (of a maximum of one). Perturbations of a thought are amplified transiently through non-normal directions while every eigenmode decays — the discrete-time analogue of hydrodynamic transient growth without spectral instability [4]. An immediate casualty is the pre-registered primary *anchor* predictor: with no eigenvalue anywhere near the unit circle, the near-unit spectral mass is identically zero and carries no signal. We report this degeneracy as-is rather than substituting a post-hoc band.

Branch prediction. The pre-registered primary branch predictor $\sigma_1(J_t)$ discriminates branch from non-branch steps only weakly (AUROC 0.554, 95% CI [0.523, 0.585]); the best local invariant

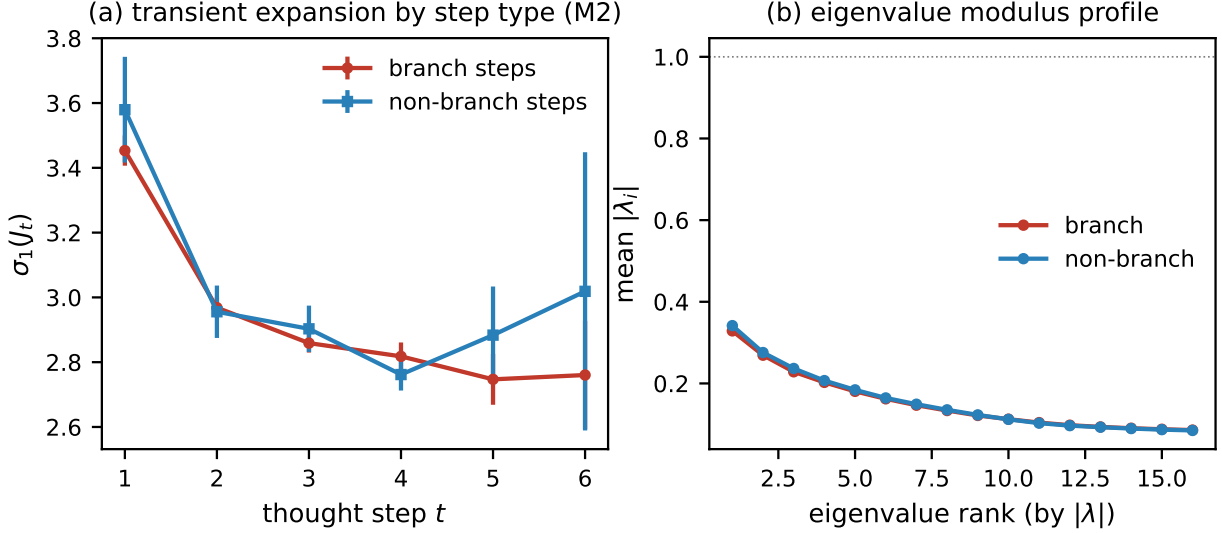


Figure 1: Spectral profile of the latent step maps (M2, 500 problems). (a) Mean leading singular value by step type: branch and non-branch steps barely differ. (b) Mean eigenvalue-modulus profile: all eigenmodes decay ($|\lambda_1| \approx 0.34$) while $\sigma_1 \approx 2.97$ — contraction with strong non-normal transient amplification.

is the non-normality ratio κ (AUROC 0.6). Both are beaten by the trivial trajectory baseline $\|c_{t+1} - c_t\|$ (AUROC 0.69).

Even the weak signal is a step-index confound. Branch steps concentrate early in the path (the root has high out-degree), and so do large trajectory jumps. When each predictor is z-scored *within* its step index before pooling — removing exactly this confound — every quantity collapses to chance: σ_1 falls to 0.492, and even the winning baseline $\|\Delta c_t\|$ falls to 0.485. Within a step position, no local quantity we measured knows which problems are at a branch point.

The null controls concur. First, the pause-token models: σ_1 computed from the identical architectural map at the identical positions predicts the same branch labels *at least as well* in the models with no latent recurrence at all — AUROC 0.556 (M3) and 0.634 (M4) vs Coconut’s 0.554 — and the per-problem branch/non-branch separation is *larger* in the nulls (ratios $3.25\times$ and $25.4\times$ Coconut’s). Second, the paired pruned-real control: deleting exactly the off-path alternatives from each problem (so every path node has out-degree one; the model still solves 97.5% of the pruned twins) does not differentially move the spectra at formerly-branch steps (mean σ_1 change -0.236 there vs -0.186 at non-branch steps; Mann-Whitney $p = 0.709$). The expansion signature does not respond to the actual presence of alternatives.

Anchors. No local invariant predicts the ablation influence I_t either: the best Spearman correlation among spectral invariants is 0.113 (CI [0.05, 0.178], the count of expanding singular directions), most other confidence intervals span zero, and the pre-registered primary (near-unit mass) is degenerate as described. The apparent top-tercile AUROCs of trajectory baselines (e.g. 0.707 for $\|\Delta c_t\|$) inherit the same step-position confound as the branch labels.

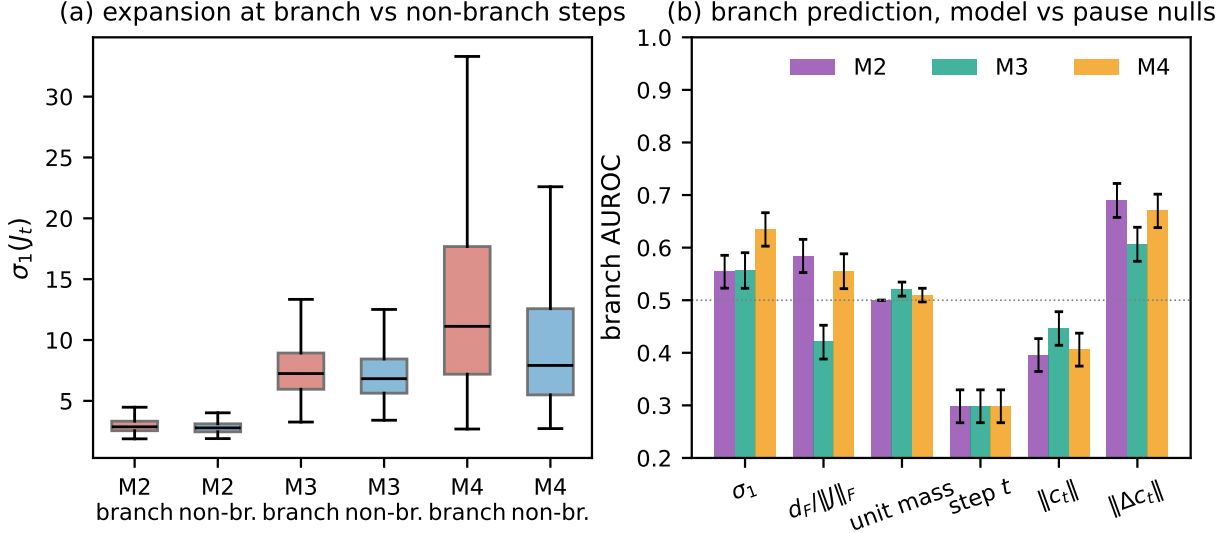


Figure 2: Branch prediction fails its nulls. (a) $\sigma_1(J_t)$ at branch vs non-branch steps for Coconut (M2) and the pause controls (M3/M4): the no-recurrence controls show *larger* separation. (b) Pooled branch AUROC with 95% cluster-bootstrap CIs for spectral invariants, the near-unit-mass diagnostic, and trivial baselines: the nulls match or beat M2 throughout.

Verdict. The pre-registered kill condition K1 fires. What weak branch- or anchor-flavored structure the local spectral invariants show is a property of the trained *architecture*, the *problem encoding*, and the *step position* — not of the latent-reasoning recurrence: it appears equally without the mechanism (pause nulls), does not respond to removing the branching itself (pruned twins), and vanishes under within-step stratification. Exactly as Aswal et al. [3] warn for geometric patterns, observable spectral patterns are not explanations.

5 E1: are the spectral directions causally load-bearing?

E0 asked whether spectral *scalars* flag reasoning structure; E1 asks the orthogonal, directional question: among perturbations of equal size, do the spectrally identified *directions* carry more causal load? For 120 problems and steps $t = 1..5$ we perturb $c_t \rightarrow c_t + s\varepsilon\|c_t\|d$ ($s = \pm 1$) with d the top right singular vector $v_1(J_t)$, the leading eigendirection $q_1(J_t)$, or magnitude-matched random unit vectors, and measure the downstream change in answer margin.

Directional privilege is real, and it is non-normal. Perturbations along $v_1(J_t)$ produce $3.32\times$ the downstream effect of random directions at $\varepsilon = 0.1$ and $7.03\times$ at $\varepsilon = 0.3$ (paired Wilcoxon $p = 1.0 \times 10^{-80}$ and $p = 1.6 \times 10^{-96}$), satisfying the pre-registered K2 criterion ($\geq 2\times$, $p < 0.01$) at both magnitudes: K2 does not fire. The leading *eigendirection*, by contrast, is not privileged ($1.09\times / 1.19\times$) — exactly the signature of non-normal dynamics, where transient growth lives in singular directions rather than eigendirections [5]. Projecting the top-4 eigen-subspace of J_t out of c_t produces $65.6\times$ the effect of removing random 4-dimensional subspaces ($p = 8.6 \times 10^{-100}$); this comparison is however not matched for removed norm, because of a striking geometric fact we report descriptively: 90.6% of each thought vector’s norm lies in the top-4 eigen-subspace of its own step Jacobian (a random 4-dimensional subspace holds 7%). The trajectory *rides* the dominant —

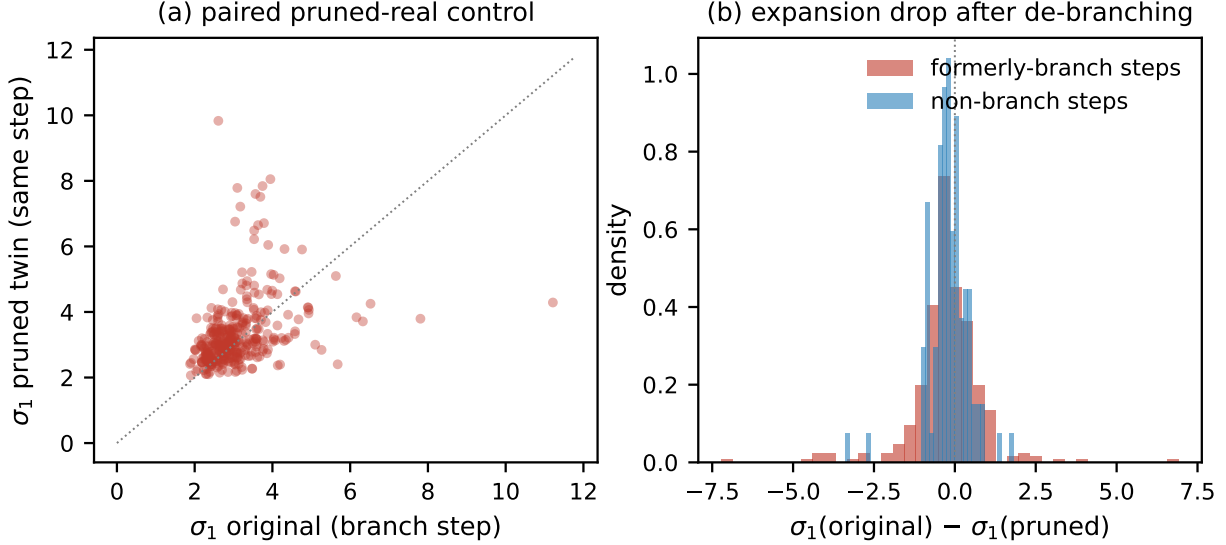


Figure 3: Paired pruned-real control. (a) σ_1 at formerly-branch steps: original problem vs its de-branched twin (identical up to deleted off-path edges; model accuracy on twins 97.5%). (b) The change distribution at formerly-branch steps matches non-branch steps (Mann–Whitney $p = 0.709$): the spectra do not respond to removing the alternatives.

yet contracting — eigenmodes of its local linearization.

But the absolute effects are strikingly small. At $\varepsilon = 0.3$ — a 30% relative perturbation — the mean margin change along v_1 is 0.054 against baseline margins of ≈ 13.9 , and *no intervention of any kind flipped a single answer* (flip rate 0% for v_1 , 0% for random). Stronger still: replacing an *entire* thought with the dataset-mean thought for its slot (the E0 ablation used as anchor ground truth) changes the margin by a median of 0.002 (90th percentile 0.008), with zero answer flips across all 500×6 single-step ablations; replacing a thought with the zero vector still flips nothing (mean $|\Delta \text{margin}| = 0.143$). In this high-accuracy Coconut checkpoint, the content of any *individual* recycled thought vector is locally, causally near-inert for the final answer: the computation is carried by the question context and the structure of the latent segment, with the residual per-thought causal signal concentrated in the non-normal amplification directions of the step map.

Post-hoc probe: the content is collectively inert. Prompted by the single-thought results, we ran one further probe, declared post-hoc in the pre-registration file before execution: replace *all six* fed thoughts simultaneously. With every thought replaced by its slot-mean, accuracy is 98.0% (baseline 98.0%) and 0 of 500 answers change. With all six thoughts *transplanted from a random donor problem*, accuracy is 98.0% and 0 answers change. Even zeroing the entire thought chain changes only 4 answers. The recycled content is not merely individually robust — it is collectively unused: what the latent segment contributes to behavior is its *structure* (six extra attended positions and whatever the cache accumulates over them), not the vectors it recycles. This is, to our knowledge, the first operator-side confirmation of the behavioral finding that the curriculum, not the recycling mechanism, drives Coconut’s ProsQA gains [8].

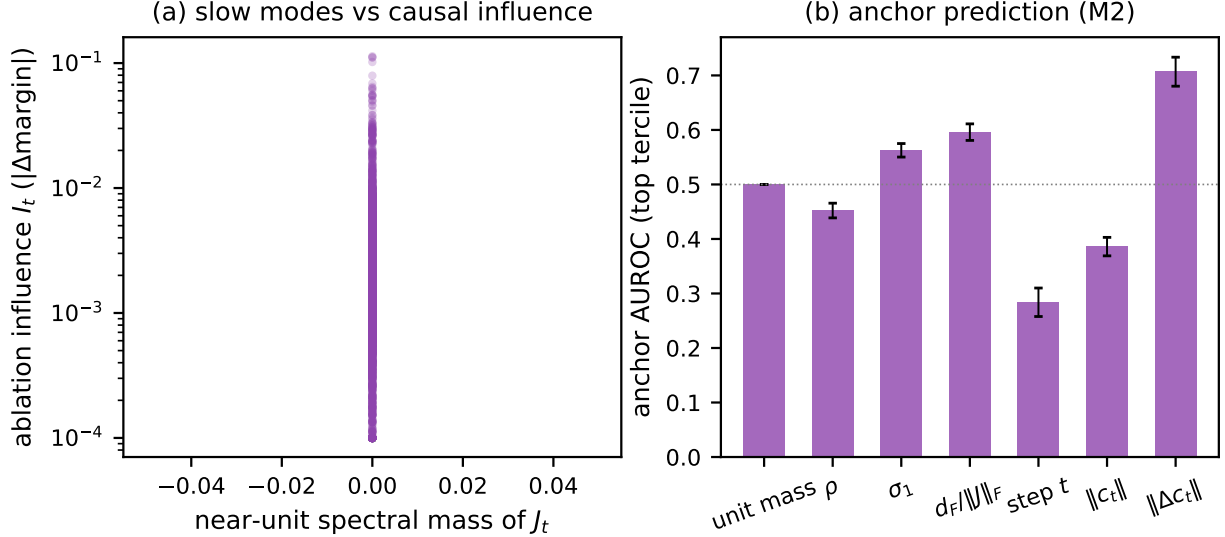


Figure 4: Anchor prediction (M2). (a) Near-unit spectral mass (pre-registered primary) vs step-ablation influence I_t : the predictor is degenerate at zero. (b) Top-tercile anchor AUROCs across invariants and baselines; apparent above-chance values carry the same step-position confound as the branch labels.

6 E2: local Jacobians, routed effects, and the orbit-level operator

A thought c_t influences later thoughts through two routes: the direct recurrence (1), whose composition the chained local Jacobians $J_{t+k-1} \cdots J_t$ describe, and the attention cache written at intermediate latent slots, which they miss. We differentiate the full unrolled map through all paths to obtain the influence Jacobian $G_{t \rightarrow t+k}$ and measure the routedness $R_{t,k} = \|G_{t \rightarrow t+k} - \prod J\|_F / \|G_{t \rightarrow t+k}\|_F$.

Half the influence is routed within three steps. At $k=2$ the local chain is a good description (median R between 0.15 and ≈ 0.15 across t), but at $k=3$ the median routedness reaches 0.51 ($t=1$), 0.48 ($t=2$), 0.44 ($t=3$): roughly *half* of the influence a thought exerts three steps ahead flows through attention rather than through the state-to-state recurrence, and the local chain systematically underestimates total influence (chain-to-full norm ratio dropping to ≈ 0.77 at $k=3$). This quantifies, in operator norms, the non-local routing that Li et al. [2] report via interventions — and it bounds what any purely state-space account of latent reasoning can capture.

The orbit-level operator does not rescue prediction. A pooled EDMD estimate over all test trajectories (PCA- $r=128$ coordinates, 99.1% variance retained) leaves a relative one-step residual of 0.359 — the pooled linear model accounts for only about two thirds of the transition — and its spectrum has 72 near-unit modes (spectral radius 1.02): persistent, context-encoding directions, as expected when each orbit is anchored to a fixed problem embedding. But its per-step diagnostics carry no reasoning-specific signal either: the Koopman one-step residual gives branch AUROC 0.554 (CI [0.521, 0.587]; the same magnitude, and the same step-position confound, as the local invariants), unit-mode participation gives 0.463, and neither correlates with ablation influence (Spearman -0.097 and -0.112). A delay-2 embedding does not change this picture.

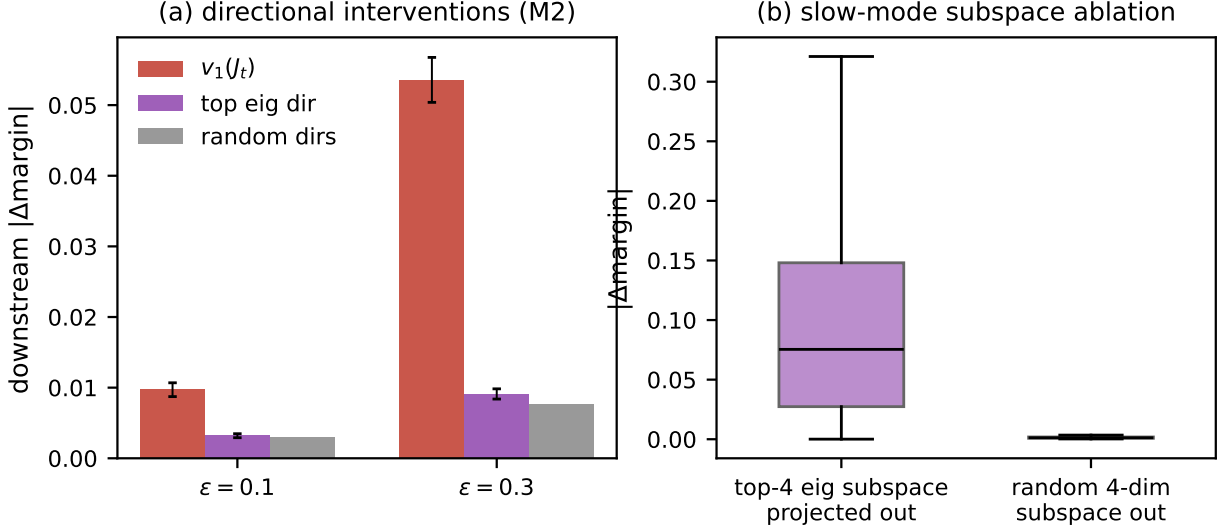


Figure 5: E1 interventions (M2, 120 problems, steps 1–5). (a) Downstream $|\Delta\text{margin}|$ for perturbations along $v_1(J_t)$, the leading eigendirection, and magnitude-matched random directions: the singular direction is privileged, the eigendirection is not. Note the absolute scale against baseline margins of ≈ 13.9 . (b) Projecting out the top-4 eigen-subspace vs random 4-dim subspaces (not norm-matched; see text — 90.6% of the thought lives in that subspace).

Verdict. The formal K3 criterion does not fire — the local σ_1 and orbit-residual branch AUROCs are statistically above chance — but the within-step stratification of Section 4 shows that this residual signal is positional rather than reasoning-specific, and we report the overall conclusion accordingly: neither the local step operator nor a pooled orbit-level transfer-operator estimate captures the anchor/branch structure of the reasoning, and the routedness measurements explain why — the latent recurrence is simply not the channel through which most influence flows.

7 E3: robustness

Held-out replication. On the 300-problem validation split (accuracy 97.3%), the σ_1 branch AUROC is 0.538 (CI [0.499, 0.579], now bracketing 0.5), and within-step stratification again collapses it — the test-split conclusions replicate out of split.

Natural linear chains. ProsQA’s generator almost never emits branch-free instances (5 in 17,886 training problems; 0 in test+valid) — the scarcity that motivated the pruned-twin construction. The model solves all 5 natural linear chains, whose mean σ_1 (3.3) is, if anything, *higher* than at genuine branch steps (3.06): further confirmation that expansion does not track branching.

The signal does not emerge with competence. Across curriculum checkpoints (Figure 7), accuracy climbs from 0% at epoch 10 to 99.0% at the best epoch while the σ_1 branch AUROC never leaves the vicinity of chance (0.461 $\rightarrow$ 0.545): the weak positional structure is not something the model *acquires* as it learns to reason.

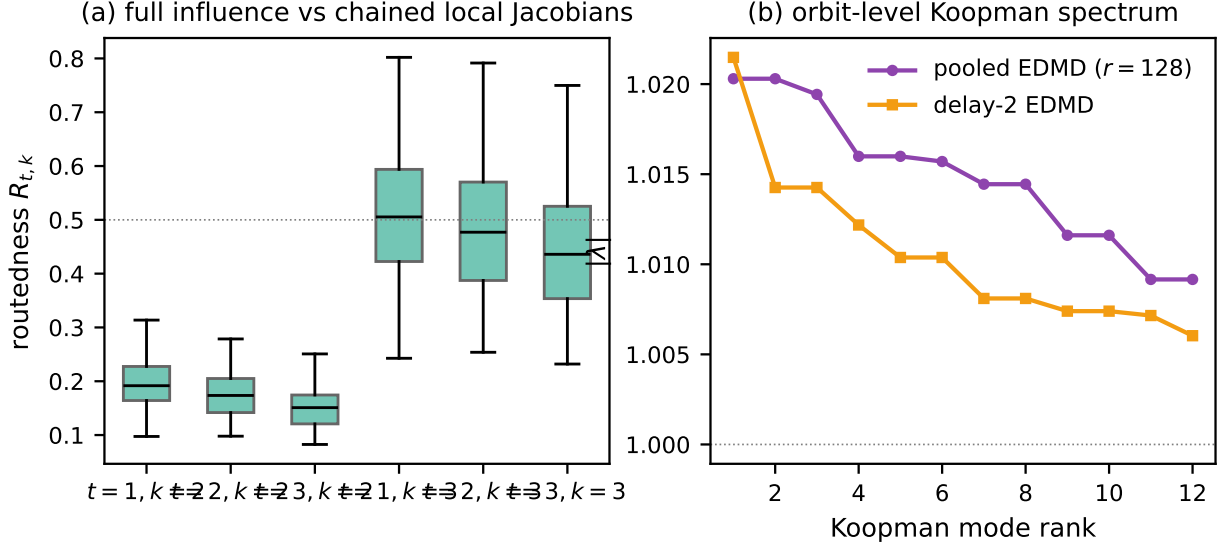


Figure 6: E2. (a) Routedness $R_{t,k}$: the fraction of the full influence Jacobian $G_{t \rightarrow t+k}$ not captured by chaining local step Jacobians rises to ≈ 0.5 at $k=3$. (b) Pooled EDMD spectrum (top mode moduli): many near-unit persistent modes — context-encoding directions — none of which carry reasoning-specific per-step signal.

Stability. All confidence intervals are stable across five bootstrap seeds (headline σ_1 CI varies by $< 10^{-3}$), and all randomized components of E1 used five seeded random-direction draws per cell.

8 Related work

Latent-reasoning interpretability. Li et al. [2] establish, via step-wise do-interventions in an SCM framing, that latent CoT has staged functionality with non-local routing — anchors exist, and influence does not propagate purely step-to-step. Aswal et al. [3] analyze Coconut and CODI against matched controls and show that causal effects concentrate in low-rank directions while observable patterns (BFS-like frontiers, decodability) appear in controls too; their methodological thesis — observable structure is not mechanism — is the discipline we adopt. Dilgren and Wiegrefe [12] find latent traces often decodable and often unnecessary. None of these works computes dynamical invariants of the thought recurrence, and none predicts anchor/branch structure ahead of measuring it; they *measure* the structure that we aim to *predict and explain*.

Spectral methods on other transformer axes. Fernando and Guitchounts [7] eigendecompose Jacobians of the residual stream treating *depth* as time in standard forward passes, finding a trained-in spectral gradient from non-normal, rotation-dominated early layers to near-symmetric late layers; this is our toolkit applied to a different dynamical object (a non-recurrent pass, no reasoning semantics). Kim et al. [6] run whitened DMD over *layer*-wise residual snapshots and show near-unit spectral mass predicts training divergence; we borrow the diagnostic, transplanted to the thought axis and an interpretability target. Carson and Reisizadeh [13] model sentence-level reasoning as an SDE with PCA+GMM regime detection — trajectory framing at the token level, without Jacobian or operator spectra and without ground-truth DAG validation. Sun et al. [14] study discrete-CoT trajectory geometry with step-specific subspaces and mid-trajectory correctness signals; no spectral

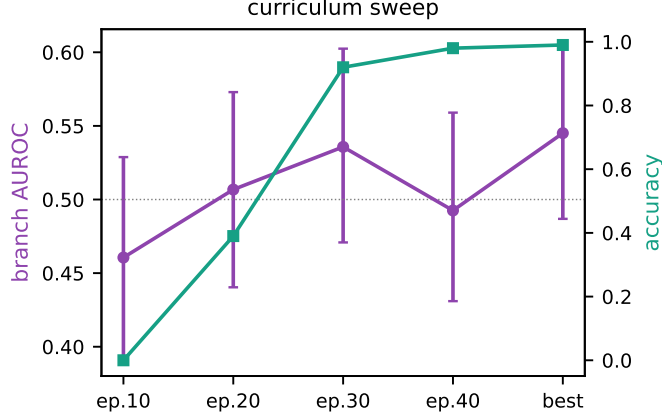


Figure 7: Curriculum sweep (100 problems per epoch checkpoint): ProsQA accuracy rises from 0% (epoch 10) to 99.0% (best) while the σ_1 branch AUROC stays flat and indistinguishable from chance throughout.

machinery. Local-Jacobian eigenspectra along recurrent trajectories have precedent in RL policy analysis [e.g. 15]; latent reasoning adds the ground-truth DAG and the matched behavioral controls that make the epiphenomenality question decidable.

Attractor-based reasoning architectures. A concurrent line *builds* fixed-point or attractor dynamics into reasoners [16–19]. We do the converse: we *analyze* the emergent dynamics of trajectories an existing latent reasoner already produces. Design and analysis are complementary; our null results, where they occur, delimit what analysis of emergent dynamics can claim.

Null-control provenance. The pause-token control precedent is Goyal et al. [10]; the specific matched suite we use — same curriculum, same positions, matched accuracy — is due to Martin [8], whose finding that the *curriculum* rather than the recycling mechanism drives Coconut’s ProsQA gains makes the pause models an unusually sharp null for any claim about the recurrence.

9 Discussion

What died, and what survived. The attractive program — read the reasoning off the spectrum of the thought recurrence — died at its pre-registered gates, in the strongest available way: not for lack of signal, but because the signal appears identically without the mechanism. What survived is a characterization with consequences. The latent step map is a contracting, strongly non-normal operator whose transient amplification is real and causally directional (v_1 privilege), whose state rides its own dominant contracting modes, and whose direct-feed channel carries strikingly little of the computation’s causal load — the answer is robust to every single-thought intervention we tried, and half of what influence there is routes through attention within three steps. In retrospect these pieces cohere: a nearly-inert, strongly-contracting recurrence is exactly the regime in which spectra can look structured while explaining nothing about behavior.

Consequences for latent-reasoning interpretability. First, the methodological point of Aswal et al. [3] extends with full force to dynamical-systems analyses: eigenvalues along a tra-

jectory are as pattern-like as decodable frontiers, and matched behavioral controls plus causal tests are the price of admission. Second, our step-position stratification deserves emphasis as a cheap, general control: in curriculum-trained latent reasoners, step index is a powerful confound that inflates *any* per-step statistic — it inflated our trivial baselines as much as our spectral predictors. Third, the routedness measurements bound the entire family of state-space accounts: any analysis that models latent reasoning as $c_{t+1} = F(c_t)$ — Koopman, SDE, attractor, or otherwise — is modeling a channel that, in this trained system, loses half its influence within three steps to a side channel it cannot see. Analyses that include the attention state (the full KV configuration) as part of the dynamical state are the natural next object.

Relation to the anchor literature. Our results do not contradict Li et al. [2]: they study different latent-CoT systems and intervention scales, and our E1 confirms that *relative* causal heterogeneity across directions is real. What we add is that in a curriculum-trained Coconut at ProsQA-scale, the per-thought causal budget is so small in absolute terms that “anchor” structure, where measurable, reflects position and architecture. The dynamical vindication of the curriculum thesis [8] — behavior preserved with the recurrence content replaced — is, to our knowledge, the first from the operator side.

10 Limitations and honest scope

Everything here is one architecture (GPT-2 124M), one training seed, one synthetic task (ProsQA) with 3–6-hop reachability, six-step latent trajectories, and one checkpoint suite. The negative results are therefore existence proofs about a representative, high-performing latent reasoner — not universal claims about latent reasoning; a system trained such that thought content is load-bearing (longer horizons, information-bottlenecked contexts, or tasks where the question cannot be re-read from the cache) could behave differently, and our harness transfers directly to such systems. Trajectories of length six bound the orbit-level analysis: pooled EDMD with a linear dictionary (plus delay-2) is the strongest operator estimate the data supports, and Ruelle–Pollicott-style resonance analysis is out of reach at this length. The pruned-twin control changes the question text (shorter prompts), though the model’s near-ceiling accuracy on the twins bounds the distribution shift. Intervention magnitudes were capped at 30% of the thought norm; larger perturbations eventually do change behavior, but at that point the intervention is no longer a local probe of the linearization. Finally, our step-ablation anchor ground truth follows the leverage protocol of prior work but is itself a single-step intervention; multi-step or subspace-level anchor definitions could reveal structure our per-step measure misses.

11 Reproducibility

All code, pre-registration (PLAN.md with kill conditions committed before results), captured results, and the scripts that regenerate every number in this paper (`paper/numbers.tex` is machine-generated; `paper/verify_regen.py` checks it) are in the project repository. Checkpoints are public (`bmarti44/coconut-curriculum-checkpoints`); data are public (`facebookresearch/coconut`). Total compute: a single RTX 3080 (10 GB).

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